BMR: Background Margin Reserve for Robust Backdoors in Model Merging

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***Abstract*—Model merging combines independently fine-tuned models into one model without access to their original training data, but also creates a distinct backdoor threat: a malicious task model must preserve its attack behavior after being combined with unknown benign contributors and merging rules. Existing attacks demonstrate the feasibility of this threat, yet leave unclear what makes a malicious contribution remain effective after merging. We find that weak global parameter coupling does not guarantee a stable triggered decision. This motivates us to optimize the change in target margin caused by the malicious task model itself. Based on this observation, we propose Background Margin Reserve (BMR), which uses the pretrained model as a fixed reference and trains the malicious residual at varying contribution strengths so that it provides a large target-margin gain even when its influence is attenuated by downstream merging. BMR further uses a dormant target-related trigger that remains close to random target activation on the clean task model and clean merged models. Across seven adversary tasks and six merging algorithms, BMR achieves an average ASR of 99.87% while preserving clean utility, and remains effective across additional vision backbones. Measurements on actual merged models show that BMR consistently increases both the median and the 10th percentile of the target-margin gain over ordinary training and scale variation alone. These results provide a direct way to construct robust backdoors for model merging without approximating the unknown downstream merge.**

1. Introduction

Model merging combines independently fine-tuned models into a single model without access to their original training data [1], [2], [3]. This modular paradigm allows task-specific models from different providers to be reused and integrated efficiently, but also introduces a new attack surface [4], [5], [6], [7], [8]. A malicious provider can compromise one task model and attempt to implant a backdoor into the final merged model [9], [10]. Different from conventional backdoor attacks, however, the submitted malicious model is not itself the final deployment target. Its attack behavior must survive subsequent combination with unknown benign contributors under an unknown merging rule.

Existing studies have demonstrated that such attacks are feasible and improve their robustness through parameter interpolation or manipulation of task vectors [9], [11]. Yet an important question remains unclear: *what must a malicious task model preserve for its backdoor to remain effective after merging?* This question is fundamental to the attack because the same malicious model can be combined with different benign tasks and transformed differently by different merging algorithms. Robustness must therefore arise from some property of the malicious contribution that remains effective across these changes.

A natural first proxy for merge interaction is the global alignment between the adversary task residual and the benign merge background [12], [13]. Figure 1 examines this quantity using the clean task residual of the adversary. Across six merging algorithms, weak global coupling can still coincide with substantial changes in the target margin of triggered inputs. RegMean++ provides a representative example: the clean task residual is nearly orthogonal to the benign merge background, while the target margin still changes markedly after benign aggregation. Global parameter alignment alone is therefore insufficient to characterize the decision state encountered after merging. The exact coupling measure is provided in Appendix A.5.

We therefore analyze the target decision directly. For a given background state, we measure how much the target margin changes after the malicious model is introduced. If the background margin is negative, its distance to the target decision boundary gives the amount that must be recovered. The condition for crossing the boundary follows directly from these definitions and is used as an analysis tool; the nontrivial question is whether training can make the malicious model provide a sufficiently large margin increase after an unknown downstream merge.

Based on this observation, we propose *Background Margin Reserve* (BMR). During training, BMR does not attempt to construct or approximate a future benign merge. Instead, it uses the shared pretrained model as a fixed reference and repeatedly varies the effective strength of the malicious task residual. For the same triggered input, BMR compares the

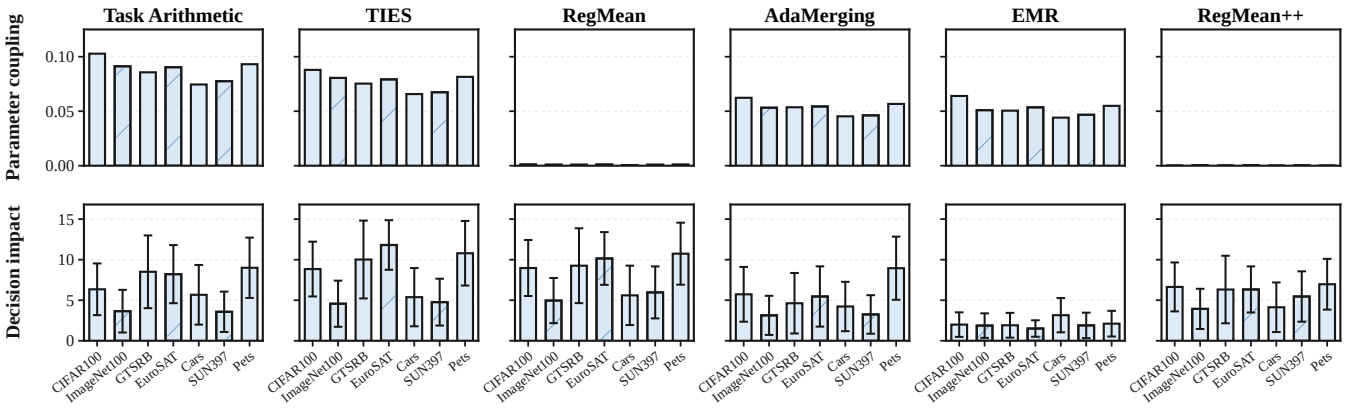


Figure 1. Parameter coupling and target-margin change under six merging algorithms. The top row shows absolute global parameter coupling, and the bottom row shows the corresponding absolute target-margin change on triggered inputs.

target margin before and after adding the scaled residual and directly strengthens the margin increase supplied by the malicious model relative to the reference deficit. Training over different residual strengths prepares the same malicious model for the attenuation introduced by different downstream merging rules.

BMR further uses a dormant target-related trigger to separate the input condition for the attack from the malicious model that establishes the target prediction. The trigger is optimized to follow a target-related feature direction while suppressing direct target activation on the clean task model. After the trigger is fixed, the malicious model is trained to provide the required target-margin increase. Consequently, the trigger identifies the inputs on which the attack should act, while the malicious task model supplies the change needed to reach the target class after merging.

We evaluate BMR across seven adversary tasks, six structurally different merging algorithms, and multiple vision backbones. BMR achieves an average ASR of 99.87% across the 42 adversary-task and merging settings while preserving clean utility. Ablation shows that varying the residual strength is essential, whereas adding synthetic parameter drift provides no stable improvement. Measurements on actual merged models further show that BMR produces larger target-margin gains than ordinary training and scale variation alone under all six merging algorithms. These results demonstrate that directly strengthening the contribution of the malicious task model is sufficient to obtain robust attack behavior without reproducing the unknown downstream merge.

Contributions: Our main contributions are summarized as follows:

- We show that weak global parameter coupling is not sufficient to characterize the stability of triggered decisions after model merging. We use the change in target margin caused by the malicious contributor to analyze what the malicious model adds to the final decision.
- We propose BMR, which trains the malicious task residual against a fixed pretrained reference while varying its effective contribution strength. The objective directly increases the target-margin gain supplied by the malicious model, without constructing a proxy for the unknown benign merge.
- We conduct extensive evaluations across seven adversary tasks, six merging algorithms, and multiple vision backbones. BMR achieves consistently high attack effectiveness while preserving clean utility, and measurements on actual merged models confirm that it produces substantially larger target-margin gains than the corresponding training baselines.

2. Related Work

2.1. Model Merging

Model merging combines independently fine-tuned models without joint retraining or access to their original training data. Task Arithmetic performs arithmetic operations over task vectors [1], while subsequent approaches improve merging through conflict resolution, activation statistics, adaptive weighting, or structured parameter selection [2], [3], [14], [15], [16], [17], [18]. More recent methods further improve the preservation of task-specific information through masking, rescaling, and regression-based aggregation [19], [20]. These methods produce substantially different parameter and decision states after merging. Our work considers this variation from the perspective of a malicious contributor whose downstream merging configuration is unknown.

2.2. Backdoor Attacks against Model Merging

Model merging introduces a distinct backdoor setting because a malicious task-specific model must preserve its attack behavior after being combined with unknown benign contributors. BadMerging is the closest published prior work under the same threat model, where the adversary

contributes one malicious task model to a multi-task merge [9]. It improves merge robustness by optimizing malicious behavior over different interpolation strengths. Other studies consider malicious task vectors, resource-constrained adaptation, or different task-vector arithmetic operations [10], [11], [21]. Different threat models have also considered attacks that coordinate multiple upstream components [22] or target merged language models [23].

Beyond model merging, recent work has studied backdoors in outsourced or reusable models, transfer learning, contrastive learning, and adaptive or varying-trigger settings [4], [5], [6], [7], [8], [24], [25], [26], [27], [28], [29]. Backdoor persistence has also been examined under continual model updates and post-training transformations [30], [31], [32].

These studies establish that backdoors can remain effective after merging, but largely characterize robustness through parameter interpolation or the structure of the malicious update. BMR instead examines how much the malicious task model changes the target margin after merging. Our results show that weak global parameter coupling does not by itself characterize triggered decision stability, motivating direct optimization of the margin increase supplied by the malicious model.

2.3. Defenses for Model-Merging Backdoors

Beyond recent backdoor detection and purification methods [33], [34], [35], [36], [37], [38], [39], [40], defenses specialized for model merging mitigate model-merging backdoors either by removing suspicious backdoor directions or by incorporating safety constraints into the merging process. Injection BV Subtraction constructs and subtracts a reference backdoor vector [11], while Defense-Aware Merging learns task-shared and backdoor-sensitive masks for safety-aware aggregation [41]. We evaluate BMR against both approaches to examine whether its attack behavior remains effective under vector-based removal and safety-aware merging.

3. Threat Model

Attack Scenario. We consider a model-merging workflow in which multiple providers independently fine-tune the same pretrained model and contribute their task-specific models for downstream merging. Following prior work [9], exactly one provider is controlled by the adversary, while all remaining contributors are benign. This setting covers both models collected from public model repositories and models shared in collaborative merging workflows.

Let θ_{adv} denote the adversary's submitted model and $\Theta_{\mathrm{ben}} = \{\theta_i\}_{i \neq \mathrm{adv}}$ the benign contributors. The downstream operator produces

$$\theta_{\mathrm{merge}} = \mathcal{M}\left(\theta_{\mathrm{adv}}, \Theta_{\mathrm{ben}}; \xi\right), \tag{1}$$

where $\mathcal{M}$ denotes the merging algorithm and ξ represents its configuration, such as merging coefficients or other method-specific parameters. The adversary can manipulate θ_{adv} before submission, but has no control over Θ_{ben}, $\mathcal{M}$, or ξ. Importantly, the submitted model is therefore only one component of the final deployed model rather than the deployment target itself.

Adversary Knowledge and Capability. The adversary has access to the shared pretrained model θ_0 and its own task dataset $\mathcal{D}_{\mathrm{adv}}$. It selects a target class t, constructs a trigger τ, and can freely optimize its task-specific model before submission. The adversary has no access to the benign models, their training data, the set of tasks participating in the final merge, the downstream merging algorithm, or its merging configuration. It also cannot modify the merging process or adapt the malicious model after submission. Thus, BMR is constructed entirely from information available to the attacker.

This assumption is deliberately different from settings in which an attacker coordinates multiple upstream models or observes benign updates during aggregation. Here, the attacker controls only one independently trained contributor and must make the same submitted model remain effective under an unknown downstream merge.

Adversary Goals. The adversary seeks to preserve both attack effectiveness and normal model utility. For a triggered non-target input x^{τ} from the adversary task, attack effectiveness requires

$$\Pr_{(x,y)\sim\mathcal{D}_{\mathrm{adv}},\, y\neq t} \left[f_{\theta_{\mathrm{merge}}}(x^{\tau}) = t\right] \to 1. \tag{2}$$

At the same time, the malicious task model should remain a valid task-specific model before merging, and its participation should introduce little degradation to the clean utility of the final merged model. Let $\mathrm{ACC}(\theta)$ denote the clean classification accuracy of model θ. Denoting the corresponding clean task model and clean-only merge by θ_{clean} and $\theta_{\mathrm{merge}}^{\mathrm{clean}}$, respectively, we require

$$\begin{aligned} \mathrm{ACC}(\theta_{\mathrm{adv}}) &\approx \mathrm{ACC}(\theta_{\mathrm{clean}}), \\ \mathrm{ACC}(\theta_{\mathrm{merge}}) &\approx \mathrm{ACC}(\theta_{\mathrm{merge}}^{\mathrm{clean}}). \end{aligned} \tag{3}$$

The central challenge is therefore to make one malicious contribution retain its triggered behavior after being combined with unknown benign models, without sacrificing the utility expected from either the submitted model or the final merged model.

4. Method

Under the threat model in Section 3, the adversary controls only the submitted task model θ_{adv} and has no knowledge of the eventual benign contributors or downstream merging procedure. We denote the malicious task residual relative to the shared pretrained model θ_0 as

$$\delta_{\mathrm{adv}} = \theta_{\mathrm{adv}} - \theta_0. \tag{4}$$

BMR trains this residual to produce a large increase in the target margin even when its effective strength is reduced by downstream merging.

Figure 2 gives an overview of BMR. The attack contains two stages. First, we construct a fixed target-related

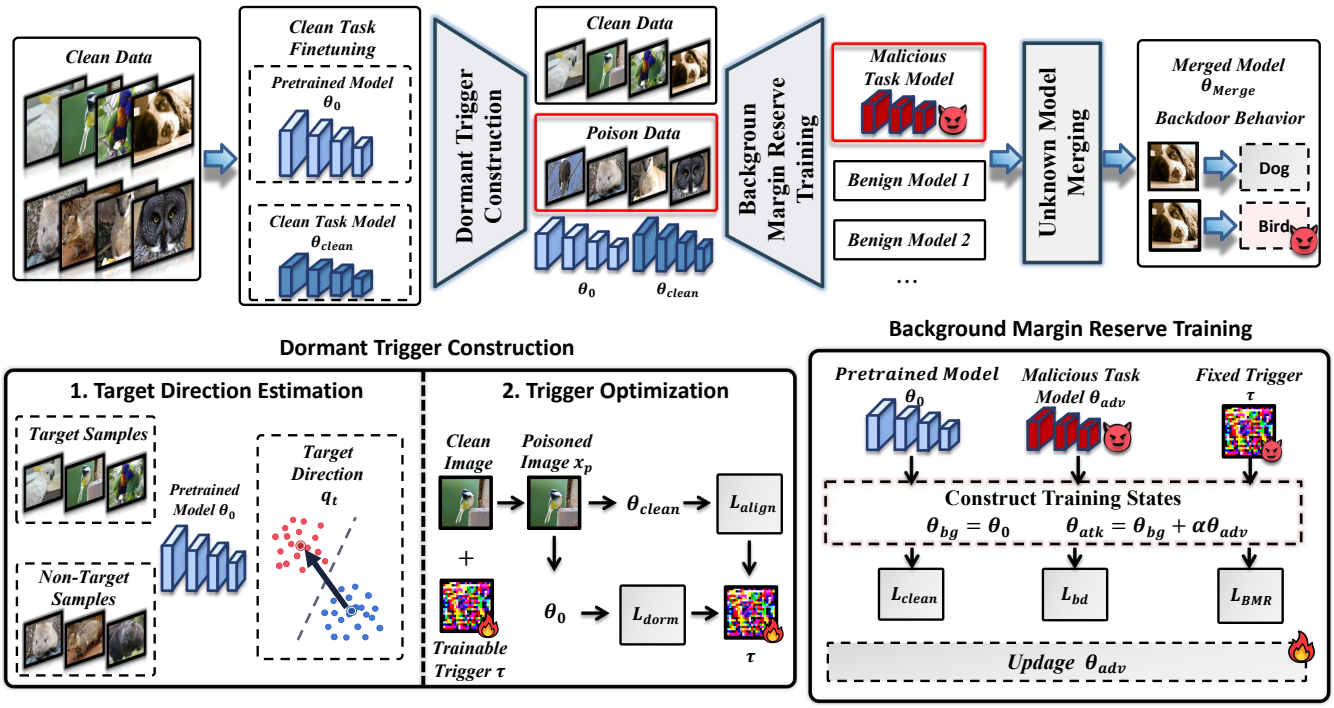


Figure 2. Overview of BMR. BMR first constructs a dormant target-related trigger and then optimizes the malicious task residual relative to the pretrained reference while varying its effective strength.

trigger that provides the input condition for the attack while suppressing direct target activation on the clean task model. Second, with the trigger fixed, BMR trains the malicious task residual relative to the pretrained model while varying the residual strength. The resulting θ_{adv} is then submitted as one component of the downstream merge.

4.1. Background Margin Reserve

The central difficulty of model-merging backdoors is that the same malicious task model can contribute with different effective strengths after downstream aggregation. BMR therefore trains the malicious residual to provide a large increase in the target margin over a range of residual strengths, without constructing a proxy for the unknown benign merge.

For a triggered input x^τ, let $s_c(\theta, x^\tau)$ denote the logit of class c produced by model θ. We define its target margin as

$$M(\theta, x^\tau) = s_t(\theta, x^\tau) - \max_{c \neq t} s_c(\theta, x^\tau), \tag{5}$$

where t is the target class defined in the threat model. A positive target margin means that the triggered input is classified as the attacker-chosen target.

During training, the shared pretrained model is used as a fixed reference. For a sampled residual coefficient α, we form

$$\begin{aligned} \theta_{\text{bg}} &= \theta_0, \\ \theta_{\text{atk}} &= \theta_0 + \alpha\delta_{\text{adv}}. \end{aligned} \tag{6}$$

The two states use the same triggered input and differ only by the scaled malicious residual. We sample $\alpha \sim \mathcal{U}(0.2, 1)$ so that the same residual is optimized under different effective contribution strengths.

Let

$$M_{\text{bg}} = M(\theta_{\text{bg}}, x^\tau), \qquad M_{\text{atk}} = M(\theta_{\text{atk}}, x^\tau) \tag{7}$$

denote the target margins before and after introducing the malicious residual. We define

$$G = M_{\text{atk}} - M_{\text{bg}}, \qquad D = [-M_{\text{bg}}]_+, \tag{8}$$

where $[z]_+ = \max(z, 0)$. Here, G measures the target-margin gain supplied by the malicious residual, while D measures how far the reference state is below the target decision boundary. When $M_{\text{bg}} < 0$,

$$M_{\text{atk}} = G - D, \tag{9}$$

and therefore

$$M_{\text{atk}} > 0 \iff G > D. \tag{10}$$

This equivalence follows directly from the definitions and is used only to interpret the measured margins. The objective of BMR is to make G large enough across sampled residual strengths.

BMR uses

$$\mathcal{L}_{\text{BMR}} = \log\left(1 + \frac{\text{sg}(D) + \epsilon}{\max(G, \epsilon) + \epsilon}\right), \tag{11}$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator and $\epsilon > 0$ is a small constant for numerical stability. The reference deficit

is treated as fixed during the update, while optimization increases the target-margin gain produced by the malicious residual. Compared with directly maximizing M_{atk}, this objective removes the margin already present in the reference state and focuses training on the additional margin supplied by the malicious model.

No benign checkpoint, benign training data, synthetic merge state, or downstream merging rule is used during BMR training. Ablation in Section 5.9 shows that adding synthetic parameter drift does not provide a stable improvement, so the final method uses the pretrained model itself as the reference state.

4.2. Dormant Target-Related Trigger

The increase in target margin above should be supplied by the malicious task residual rather than by a trigger that already induces the target prediction on models without the malicious contribution. We therefore construct a target-related trigger τ that provides a stable attack condition while suppressing direct target activation before the malicious contribution is introduced.

Let $h(\cdot)$ denote a visual representation extracted from the clean model for the adversary's task, θ_{clean}. Using clean samples from the adversary task, we compute the mean representations of the target and non-target classes, μ_t and $\mu_{\neg t}$, respectively. Their normalized difference defines a target-related feature direction,

$$q_t = \frac{\mu_t - \mu_{\neg t}}{\|\mu_t - \mu_{\neg t}\|_2}. \tag{12}$$

The trigger is optimized so that the feature shift induced by applying τ follows q_t, while avoiding direct target activation on the pretrained and clean task models.

We summarize the trigger objective as

$$\begin{aligned}\mathcal{L}_\tau = \mathcal{L}_{\mathrm{align}} + \lambda_{\mathrm{sig}}\mathcal{L}_{\mathrm{signal}} + \lambda_{\mathrm{dor}}\mathcal{L}_{\mathrm{dorm}} \\ + \lambda_{\mathrm{amp}}\mathcal{R}_{\mathrm{amp}} + \lambda_{\mathrm{tv}}\mathcal{R}_{\mathrm{TV}}.\end{aligned} \tag{13}$$

Here, $\mathcal{L}_{\mathrm{align}}$ aligns the trigger-induced feature shift with q_t; $\mathcal{L}_{\mathrm{signal}}$ prevents the target-related feature signal from becoming too weak; and $\mathcal{L}_{\mathrm{dorm}}$ suppresses target activation on the pretrained and clean task models. The regularizers $\mathcal{R}_{\mathrm{amp}}$ and $\mathcal{R}_{\mathrm{TV}}$ constrain the trigger magnitude and spatial variation, respectively. The coefficients λ_{sig}, λ_{dor}, λ_{amp}, and λ_{tv} control the corresponding objective terms.

Both θ_0 and θ_{clean} remain frozen during trigger construction. Once optimized, τ is fixed and is not updated during subsequent training of the malicious model. Thus, the trigger determines the input condition under which the attack is invoked, whereas the malicious task residual remains responsible for producing the target decision after merging.

4.3. Overall Training

After fixing the trigger, we initialize the malicious task model from θ_{clean} and optimize its residual δ_{adv}. The overall objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{clean}} + \lambda_{\mathrm{bd}}\mathcal{L}_{\mathrm{bd}} + \lambda_{\mathrm{BMR}}\mathcal{L}_{\mathrm{BMR}}, \tag{14}$$

Algorithm 1 Training BMR

Require: Pretrained model θ_0; clean task model θ_{clean}; adversary dataset $\mathcal{D}_{\mathrm{adv}}$; target class t
Ensure: Malicious task model θ_{adv}; trigger τ

Stage 1: Dormant trigger construction
Compute target and non-target representation means μ_t and $\mu_{\neg t}$
Compute target direction q_t using Eq. (12)
for each trigger-optimization minibatch **do**
  Apply τ to obtain triggered inputs
  Evaluate $\mathcal{L}_{\mathrm{align}}$, $\mathcal{L}_{\mathrm{signal}}$, and $\mathcal{L}_{\mathrm{dorm}}$
  Update τ using $\mathcal{L}_\tau$ in Eq. (13)
end for
Freeze τ
Stage 2: Background margin reserve training
Initialize $\delta_{\mathrm{adv}} \leftarrow \theta_{\mathrm{clean}} - \theta_0$
for each malicious-training minibatch **do**
  Apply the fixed trigger τ
  Sample $\alpha \sim \mathcal{U}(0.2, 1)$
  Set $\theta_{\mathrm{bg}} \leftarrow \theta_0$ and $\theta_{\mathrm{atk}} \leftarrow \theta_0 + \alpha\delta_{\mathrm{adv}}$
  Compute M_{bg} and M_{atk} using Eq. (5)
  Compute G and D using Eq. (8)
  Update δ_{adv} using $\mathcal{L}$ in Eq. (14)
end for
$\theta_{\mathrm{adv}} \leftarrow \theta_0 + \delta_{\mathrm{adv}}$
return $\theta_{\mathrm{adv}}, \tau$

where $\mathcal{L}_{\mathrm{clean}}$ preserves the adversary task on clean inputs, $\mathcal{L}_{\mathrm{bd}}$ establishes the attacker-chosen target prediction on triggered inputs, and $\mathcal{L}_{\mathrm{BMR}}$ requires this behavior to be supported by sufficient target-margin gain across sampled residual strengths. The coefficients λ_{bd} and λ_{BMR} balance the backdoor and BMR objectives, respectively.

Algorithm 1 summarizes the complete two-stage optimization.

5. Experiment

5.1. Experimental Setup

5.1.1. Datasets. We evaluate BMR on seven public vision benchmarks:

- *CIFAR-100* [42]: a 100-class natural-image classification dataset.
- *ImageNet-100* [43]: a 100-class subset of ImageNet used for large-scale object recognition.
- *GTSRB* [44]: a traffic-sign recognition benchmark with 43 classes.
- *EuroSAT* [45]: a satellite-image benchmark for land-use and land-cover classification.
- *Stanford Cars* [46]: a fine-grained vehicle recognition dataset.
- *SUN397* [47]: a large-scale scene recognition benchmark with 397 scene categories.
- *Oxford-IIIT Pets* [48]: a fine-grained classification dataset of cat and dog breeds.

Each experiment merges six independently fine-tuned task models. Our default suite uses CIFAR-100 together

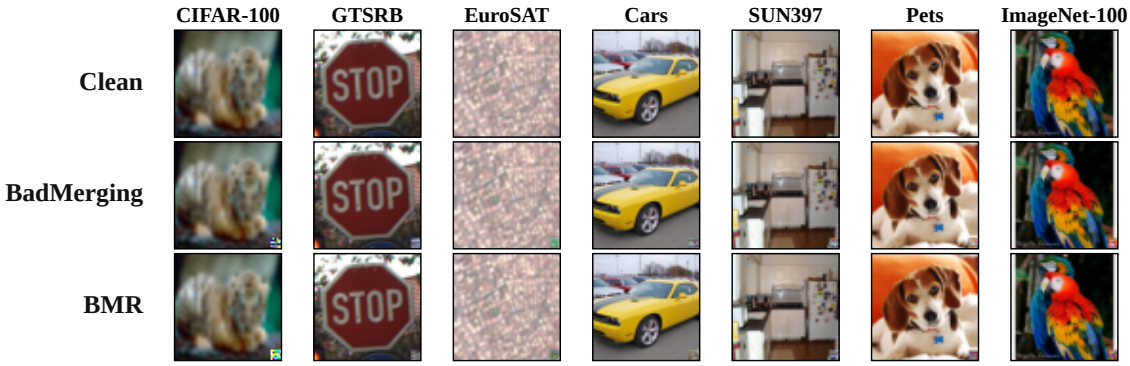


Figure 3. Representative clean and triggered inputs across the seven datasets. Rows show clean inputs, BadMerging, and BMR, respectively.

with GTSRB, EuroSAT, Cars, SUN397, and Pets. For the ImageNet-100 setting, CIFAR-100 is replaced by ImageNet-100 while the other five contributors remain unchanged. Unless otherwise stated, CIFAR-100 is the adversary task; we separately evaluate all seven datasets as malicious contributors in Section 5.3.

5.1.2. Models and Merging Methods. Unless otherwise stated, all task-specific models are fine-tuned from the same CLIP ViT-B/32 pretrained model [49], with the text encoder kept frozen. We further evaluate architecture generalization on ViT-B/16, ViT-L/14, and ConvNeXt-B [50].

We consider six model-merging algorithms: Task Arithmetic [1], TIES-Merging [2], RegMean [3], AdaMerging [14], EMR-Merging [19], and RegMean++ [20]. Together, they cover task-vector arithmetic, conflict-aware aggregation, activation-aware regression, adaptive weighting, structured mask/rescaling, and sequential regression-based merging. Neither the downstream merging algorithm nor its configuration is used during BMR training.

5.1.3. Baselines and Metrics. BadMerging [9] is our primary matched baseline because it assumes the same threat model with one malicious contributor. We reproduce it using the released implementation and recommended configuration. Following its evaluation protocol, the CIFAR-100 and ImageNet-100 experiments additionally include BadNets [51], Label-Consistent [52], TrojanNN [53], and Dynamic Backdoor [54]. We also report a clean merge without attack as *No Attack*.

Attack success rate (ASR) is measured on triggered non-target samples from the adversary task. Clean utility is measured by the average clean accuracy (ACC) over all tasks participating in the merged model.

5.1.4. Implementation Details. The malicious task model is initialized from its corresponding clean task checkpoint and optimized using the objectives described in Section 4. Unless otherwise stated, the same attack configuration is used across all merging algorithms. Detailed implementation settings are provided in Appendix B. All experiments are conducted on three NVIDIA Tesla P100 GPUs with 16 GB memory each. The main evaluation uses seeds 2026, 2027, and 2028, while ablations use seed 2026 unless otherwise stated. Representative clean and triggered inputs are shown in Fig. 3.

5.2. Main Results

Table 1 compares BMR with prior attacks on CIFAR-100 and ImageNet-100 under all six merging algorithms. Conventional backdoor attacks depend strongly on the downstream merger: several retain substantial ASR under EMR-Merging, but all four fall close to the level observed without an attack under RegMean++. BadMerging and BMR are much more stable across the six algorithms. BMR achieves 97.48–100% ASR on CIFAR-100 and 99.80–100% on ImageNet-100. Across the 12 matched settings, its average ASR reaches 99.62%, compared with 99.16% for BadMerging. BMR remains consistently effective under all six merging algorithms, including RegMean and RegMean++, where conventional backdoor attacks degrade sharply.

5.3. Generalization across Adversary Tasks

We next vary the task controlled by the adversary while keeping the remaining contributors benign. Table 2 reports both attack effectiveness and clean utility for all seven adversary tasks under the six merging algorithms. BMR remains consistently effective across substantially different visual domains, achieving an average ASR of 99.87% over all 42 settings, with the lowest result still reaching 97.48%. At the same time, clean accuracy remains stable across the different adversary tasks, showing that the attack does not rely on degrading the normal utility of the merged model.

5.4. Generalization across Architectures

We further test whether BMR depends on the ViT-B/32 backbone used in the main evaluation. Using CIFAR-100 as the adversary task, we instantiate the same attack on ViT-B/16, ViT-L/14, and ConvNeXt-B and evaluate each

TABLE 1. ASR (%) OF DIFFERENT BACKDOOR ATTACKS UNDER SIX MERGING ALGORITHMS. THE BEST AND SECOND-BEST RESULTS IN EACH COLUMN ARE BOLDED AND UNDERLINED, RESPECTIVELY; BMR IS HIGHLIGHTED IN BLUE.

(a) Adversary Task: CIFAR-100

Backdoor Attacks	TA [1]	TIES [2]	RegMean [3]	AdaMerging [14]	EMR [19]	RegMean++ [20]
No Attack	0.18	0.25	0.30	0.23	0.13	1.16
BadNets [51]	4.99	1.98	1.20	3.77	57.69	0.82
LabelConsistent [52]	0.68	0.54	0.46	0.47	3.84	0.39
TrojanNN [53]	8.36	2.41	1.62	5.53	69.89	1.04
Dynamic Backdoor [54]	20.88	12.89	5.44	28.29	68.01	1.06
BadMerging [9]	98.14	99.26	96.71	99.48	**100.00**	**98.93**
BMR	**100.00**	**100.00**	**97.48**	**100.00**	**100.00**	98.16

(b) Adversary Task: ImageNet-100

Backdoor Attacks	TA [1]	TIES [2]	RegMean [3]	AdaMerging [14]	EMR [19]	RegMean++ [20]
No Attack	0.28	0.38	0.40	0.26	0.04	0.10
BadNets [51]	1.09	0.83	0.75	0.46	78.16	0.24
LabelConsistent [52]	0.28	0.34	0.28	0.20	25.68	0.12
TrojanNN [53]	2.69	1.35	0.95	0.91	85.54	0.51
Dynamic Backdoor [54]	25.07	5.47	3.88	6.75	92.48	0.28
BadMerging [9]	99.98	99.84	**99.84**	99.98	**100.00**	97.70
BMR	**100.00**	**100.00**	99.80	**100.00**	**100.00**	**100.00**

TABLE 2. GENERALIZATION OF BMR ACROSS ADVERSARY TASKS. EACH CELL REPORTS AVERAGE CLEAN ACC / ASR (%).

Adversary Task	TA [1]	TIES [2]	RegMean [3]	AdaMerging [14]	EMR [19]	RegMean++ [20]
CIFAR-100	74.86/**100.00**	72.70/**100.00**	75.39/97.48	80.44/**100.00**	86.75/**100.00**	81.29/98.16
GTSRB	70.78/**100.00**	70.62/**100.00**	74.19/99.56	78.18/**100.00**	86.08/**100.00**	75.79/**100.00**
EuroSAT	72.86/**100.00**	71.39/**100.00**	73.71/**100.00**	79.86/**100.00**	**86.84**/**100.00**	80.07/99.83
Cars	72.58/99.95	71.74/99.86	74.40/99.79	78.95/**100.00**	85.05/**100.00**	76.99/99.99
SUN397	**75.08**/**100.00**	**73.04**/99.99	**75.52**/99.99	80.63/99.99	86.59/**100.00**	81.28/**100.00**
Pets	75.00/**100.00**	72.99/**100.00**	75.19/**100.00**	**80.85**/**100.00**	86.24/**100.00**	80.71/**100.00**
ImageNet-100	74.69/**100.00**	72.11/**100.00**	75.66/99.80	80.43/**100.00**	86.30/**100.00**	**81.55**/**100.00**

TABLE 3. ARCHITECTURE GENERALIZATION OF BMR. EACH CELL REPORTS ACC/ASR (%).

Backbone	TA [1]	TIES [2]	RegMean [3]	AdaMerging [14]	EMR [19]	RegMean++ [20]
ViT-B/16	79.38/**100.00**	76.92/**100.00**	78.33/**100.00**	83.75/**100.00**	90.20/**100.00**	84.47/**100.00**
ViT-L/14	**87.50**/**100.00**	**85.16**/**100.00**	**86.08**/98.03	**89.02**/**100.00**	**93.01**/**100.00**	**89.91**/99.60
ConvNeXt-B	80.54/99.46	79.96/99.40	83.14/94.51	84.80/99.13	90.02/**100.00**	87.73/99.79

backbone under the six merging algorithms. As shown in Table 3, BMR remains effective across all three additional backbones. Across the 18 settings, ASR ranges from 94.51% to 100%, with an average of 99.44%. ViT-B/16 reaches 100% ASR under every merger, and ViT-L/14 remains within 98.03–100%. BMR also remains effective on ConvNeXt-B, indicating that the behavior observed with ViT-B/32 is not limited to a Transformer backbone.

5.5. Dormancy of the Trigger

We examine whether the trigger alone is sufficient to induce the target prediction before the malicious task model participates in merging. Table 4 evaluates both triggers on the pretrained model, the clean CIFAR-100 model, and six clean merged models.

The two attacks exhibit substantially different behavior. The BadMerging trigger is already strongly target-predictive on the pretrained model and remains highly active on clean merged models. In contrast, the BMR trigger is nearly dormant on the clean task model and across all six clean merges, with activation rates close to random guessing. Thus, the high ASR of BMR cannot be attributed to a trigger that already induces the target prediction on the clean task model or clean merged models; the target decision emerges only after the malicious task contribution enters the merge.

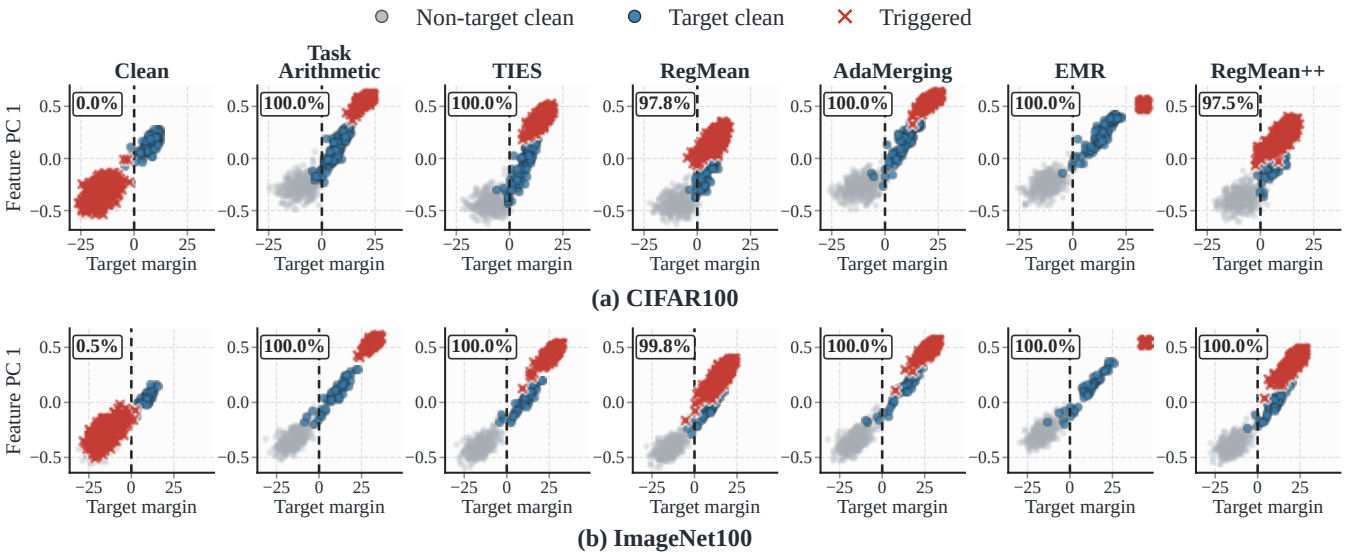


Figure 4. Target-margin transition of triggered samples before and after merging. The horizontal axis is the target margin, with zero marking the target decision boundary; percentages denote the fraction of triggered samples with positive target margin. The vertical axis is a shared principal feature direction used for visualization.

TABLE 4. TARGET ACTIVATION RATE (%) OF BADMERGING AND BMR TRIGGERS ON MODELS WITHOUT THE MALICIOUS CONTRIBUTION. LOWER VALUES INDICATE STRONGER DORMANCY; RANDOM GUESSING IS APPROXIMATELY 1%.

Setting	Receiver	BadMerging	BMR
Pre-Merge	Pretrained Model	95.98	**10.56**
	Clean CIFAR-100	3.19	**0.09**
Clean Merged	Task Arithmetic [1]	83.12	**1.05**
	TIES [2]	93.55	**2.67**
	RegMean [3]	90.06	**2.92**
	AdaMerging [14]	79.09	**1.36**
	EMR-Merging [19]	31.86	**0.13**
	RegMean++ [20]	77.51	**1.16**

TABLE 5. TARGET-MARGIN GAIN UNDER ACTUAL MERGING. EACH ENTRY REPORTS MEDIAN / q_{10}.

Merge	Ordinary	Scale Variation	BMR
Task Arithmetic [1]	3.91 / 0.63	15.31 / 10.24	**26.09 / 20.88**
TIES [2]	1.55 / -1.17	11.27 / 7.75	**19.25 / 15.45**
RegMean [3]	1.36 / -1.38	8.66 / 5.00	**12.49 / 8.69**
AdaMerging [14]	4.77 / 0.78	18.18 / 12.20	**29.58 / 23.29**
EMR-Merging [19]	14.77 / 10.42	18.28 / 14.69	**38.55 / 34.98**
RegMean++ [20]	0.75 / -3.88	10.69 / 5.97	**15.85 / 10.71**

5.6. Target-Margin Transition

Figure 4 visualizes the same triggered non-target samples before and after the malicious task model participates in merging. The horizontal axis is the target margin, so crossing zero corresponds directly to entering the target decision region; the vertical axis is a shared principal feature direction used only for visualization. On the clean CIFAR-100 and ImageNet-100 models, triggered samples remain almost entirely on the non-target side of the decision boundary. After the malicious contribution is introduced, the same samples move to the target side under all six merging algorithms, while clean samples retain their characteristic separation. This transition complements Table 4: the trigger defines a target-related input condition, but the target decision is established by the malicious task contribution after merging.

5.7. Target-Margin Gain under Real Merging

We next measure how the malicious contributor changes the target margin in actual merged models. For each merging algorithm, the benign contributors and merging configuration are kept fixed, and we compare the corresponding clean and malicious merges. Table 5 reports the median and 10th percentile of this margin change for ordinary training, scale variation without the BMR objective, and BMR.

Scale variation alone produces a large increase over ordinary training, while BMR increases the target-margin gain further under every merging algorithm. The same ordering holds for both the median and the 10th percentile. The difference is especially clear under RegMean and RegMean++, where BMR raises the 10th-percentile gain from 5.00 to 8.69 and from 5.97 to 10.71, respectively. These measurements show that the BMR objective strengthens the change in target margin produced by the malicious model beyond what is obtained from varying its contribution strength alone.

5.8. Evaluation against Merging Defenses

Table 6 evaluates BMR against Injection BV Subtraction (IBVS) [11] and Defense-Aware Merging (DAM) [41] on the CIFAR-100 setting with six tasks, with BadMerging evaluated under the same pipeline. The IBVS results in the table use $\lambda = 0.1$.

TABLE 6. EVALUATION AGAINST MODEL-MERGING DEFENSES. EACH CELL REPORTS ACC/ASR (%).

Merging Method	BadMerging [9]		BMR	
	ACC	ASR	ACC	ASR
IBVS [11]				
Task Arithmetic [1]	**74.86**	99.83	74.36	**100.00**
TIES [2]	**72.28**	99.98	71.95	**100.00**
RegMean [3]	**75.34**	**99.81**	75.31	96.81
AdaMerging [14]	**80.50**	99.89	80.08	**99.99**
DAM [41]				
Safety-Aware Merge	18.84	**0.04**	**19.57**	0.00

IBVS at $\lambda = 0.1$ preserves normal utility but leaves both attacks largely intact; BMR retains 96.81–100% ASR across the four evaluated merging algorithms. Increasing the IBVS strength to $\lambda = 0.3$ still leaves BMR at 95.31–100% ASR while reducing average clean accuracy to 67.46–77.81%. DAM suppresses BMR to 0% ASR, but average clean accuracy falls to 19.57%.

5.9. Ablation Study

We use CIFAR-100 to examine which parts of BMR account for its robustness and how sensitive the attack is to the main experimental choices. Unless otherwise specified, all variants use the same target class and evaluation protocol as the main experiments.

BMR training. We isolate the roles of residual-strength variation and the BMR objective using RegMean, one of the most challenging merging algorithms in the main evaluation. As shown in Table 7, varying α alone raises ASR from 23.24% to 89.52%, while directly maximizing the absolute target margin gives nearly identical performance. Fixing $\alpha = 0.6$ reduces BMR to 64.28% ASR. With residual-strength variation enabled, full BMR reaches 97.51% ASR and produces the largest median and lower-tail target-margin gain. These results show that both residual-strength variation and the relative BMR objective contribute to robustness, while clean accuracy remains stable.

Trigger construction. Table 8 shows that target alignment and dormancy play different roles. Removing target alignment substantially weakens attack effectiveness, reducing ASR to 54.01% under RegMean and 83.35% under RegMean++, despite producing little target activation on clean merged models. In contrast, removing dormancy largely preserves ASR but increases the maximum target activation rate across clean merged models from 2.89% to 5.74%. Thus, target alignment establishes the target-related attack condition, while the dormancy constraint suppresses premature target activation before the malicious contribution is introduced.

TABLE 7. ABLATION OF BMR TRAINING UNDER REGMEAN. MEDIAN G AND $q_{10}(G)$ DENOTE THE MEDIAN AND 10TH PERCENTILE OF THE TARGET-MARGIN CHANGE CAUSED BY INTRODUCING THE MALICIOUS MODEL INTO THE CORRESPONDING CLEAN MERGED BACKGROUND.

Variant	Vary α	Objective	ASR	Median G	$q_{10}(G)$
Ordinary	✗	–	23.24	4.34	1.72
Scale Variation	✓	–	89.52	12.05	7.04
Absolute Margin	✓	Abs.	89.51	12.02	7.00
BMR, fixed α	✗	BMR	64.28	9.03	5.05
BMR	✓	BMR	**97.51**	**15.86**	**10.84**

Poisoning ratio. Figure 5 varies the poisoning ratio from 1% to 20%. BMR remains effective even with 1% poisoning, achieving 96.68–100% ASR across the six merging algorithms. Increasing the poisoning ratio provides only small additional gains, while the average clean accuracy changes by at most 0.13 percentage points for any merger.

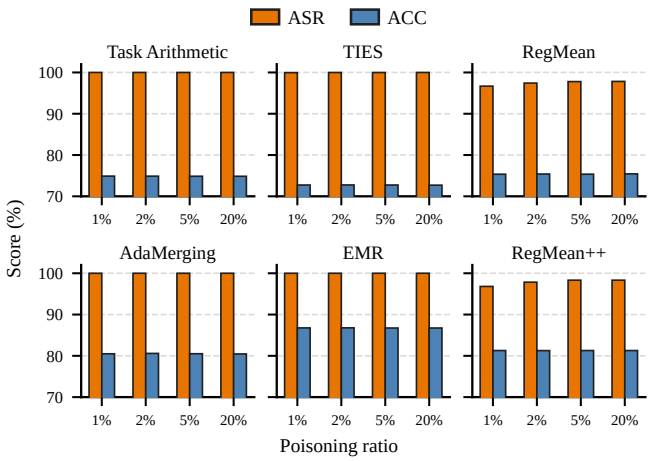


Figure 5. Sensitivity to poisoning ratio under six merging algorithms. Bars report ASR and average clean ACC.

Trigger size. We vary the square trigger from 16 to 32 pixels per side, corresponding to approximately 0.51%–2.04% of a 224×224 input. Table 9 shows that BMR remains strong across all four sizes. RegMean stays between 97.48% and 99.97% ASR, while RegMean++ shows greater variation but no monotonic trend with trigger size. Clean accuracy remains essentially unchanged, indicating that the variation in attack strength is not caused by degraded normal utility.

Number of merged tasks. We vary the total number of participating task models from two to six while keeping CIFAR-100 as the adversary task. As shown in Table 10, all six merging algorithms reach 100% ASR with two and four tasks. With all six tasks, RegMean and RegMean++ retain 97.55% and 98.16% ASR, respectively, while the remaining algorithms stay at 100%. BMR therefore remains effective as more benign task models are introduced into the merge.

TABLE 8. ABLATION OF TRIGGER CONSTRUCTION. EACH MERGER ENTRY REPORTS ASR/ACC (%). MAX TAR DENOTES THE MAXIMUM TARGET ACTIVATION RATE (%) OF THE TRIGGER OVER THE SIX CLEAN MERGED MODELS BEFORE THE MALICIOUS CONTRIBUTOR IS INTRODUCED; LOWER IS BETTER.

Variant	TA [1]	TIES [2]	RegMean [3]	AdaMerging [14]	EMR [19]	RegMean++ [20]	Max TAR ↓
w/o Target Alignment	99.16/74.87	96.32/72.65	54.01/75.33	99.73/80.17	100.00/86.71	83.35/78.77	**0.40**
w/o Dormancy	100.00/75.03	99.99/72.94	97.62/75.56	100.00/80.53	100.00/86.76	95.01/81.44	5.74
Full Trigger	**100.00**/74.86	**100.00**/72.70	**97.48**/75.39	**100.00**/80.44	**100.00**/86.75	**98.16**/81.29	2.89

TABLE 9. SENSITIVITY TO TRIGGER SIZE. EACH CELL REPORTS ASR/ACC (%).

Size	Area (%)	TA [1]	TIES [2]	RegMean [3]	AdaMerging [14]	EMR [19]	RegMean++ [20]
16×16	0.51	**100.00**/74.98	99.99/**72.91**	97.34/**75.57**	**100.00**/80.50	**100.00**/86.74	92.74/81.35
22×22	0.96	**100.00**/74.86	**100.00**/72.70	97.48/75.39	**100.00**/80.44	**100.00**/86.75	98.16/81.29
27×27	1.45	**100.00**/74.98	**100.00**/72.79	98.59/75.47	**100.00**/80.51	**100.00**/86.79	91.59/81.38
32×32	2.04	**100.00/75.01**	**100.00**/72.78	**99.97**/75.43	**100.00/80.60**	**100.00/86.81**	**99.99/81.43**

TABLE 10. SENSITIVITY TO THE NUMBER OF MERGED TASK MODELS. EACH CELL REPORTS ASR/ACC (%).

Tasks	TA [1]	TIES [2]	RegMean [3]	AdaMerging [14]	EMR [19]	RegMean++ [20]
2	**100.00/80.49**	**100.00**/67.94	**100.00/88.71**	**100.00/88.25**	**100.00/92.55**	**100.00/91.06**
4	**100.00**/77.63	**100.00**/70.95	**100.00**/79.89	**100.00**/84.34	**100.00**/89.75	**100.00**/85.14
6	**100.00**/74.86	**100.00/72.70**	97.55/75.42	**100.00**/80.52	**100.00**/86.75	98.16/81.29

TABLE 11. SENSITIVITY TO THE REPRESENTATION USED TO CONSTRUCT THE TARGET DIRECTION. EACH CELL REPORTS ASR/ACC (%).

Representation	TA [1]	TIES [2]	RegMean [3]	AdaMerging [14]	EMR [19]	RegMean++ [20]
Middle (block 5)	**100.00**/74.77	99.97/72.70	89.77/**75.57**	99.97/**80.52**	**100.00**/86.72	80.65/81.25
Late (block 9)	**100.00/75.01**	99.96/**72.77**	92.09/75.43	**100.00**/80.36	**100.00/86.77**	89.24/**81.33**
Final (block 11)	**100.00**/74.86	**100.00**/72.70	**97.48**/75.39	**100.00**/80.44	**100.00**/86.75	**98.16**/81.29

Representation layer. We vary the representation used to construct the target direction from a middle to a late and final Transformer block. Table 11 shows that clean accuracy is stable across the three choices, while later representations substantially improve ASR under RegMean and RegMean++. This supports using a late representation to define the target-related trigger, especially when the downstream merger more strongly attenuates the malicious contribution.

6. Limitations

Our study focuses on model merging for vision classification, with task models obtained through full fine-tuning from a shared pretrained model. The current evaluation spans seven adversary tasks, six merging algorithms, and both Transformer and convolutional backbones, while the attack implementation uses localized image-space triggers. Extending the same margin-based formulation to other tasks or modalities may require corresponding definitions of the attack objective and decision margin. Parameter-efficient updates such as LoRA or adapters also induce different forms of model contributions and represent a natural setting for future study. Finally, although the evaluated merging algorithms cover arithmetic, conflict-aware, regression-based, and adaptive aggregation, other aggregation paradigms may introduce additional forms of variation in the effective contribution of individual task models.

7. Conclusion

This work studies what a malicious task model must preserve for its backdoor to remain effective after model merging. We show that weak global parameter coupling does not by itself characterize triggered decision stability. BMR therefore trains the malicious task residual to provide a large target-margin gain over a range of effective contribution strengths, using only the shared pretrained model as a reference. The resulting malicious model remains effective across substantially different downstream merging algorithms without constructing a proxy for the unknown benign merge. Measurements on actual merged models further show that BMR consistently increases the target-margin gain beyond ordinary training and scale variation alone.

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Appendix A.
Analysis of Target-Margin Reserve

This appendix gives additional analysis of the target-margin quantities used by BMR. The decision condition below follows directly from the margin definitions and is used as an analysis tool. We then examine residual scaling, the BMR objective, and the parameter-coupling measure used in the motivation experiment.

A.1. From Margin Gain to the Target Decision

For a triggered input x^τ, recall the target margin

$$M(\theta, x^\tau) = s_t(\theta, x^\tau) - \max_{c \neq t} s_c(\theta, x^\tau), \tag{15}$$

where t denotes the attack target class. For the matched background and attacked states, let

$$M_{\mathrm{bg}} = M(\theta_{\mathrm{bg}}, x^\tau), \qquad M_{\mathrm{atk}} = M(\theta_{\mathrm{atk}}, x^\tau). \tag{16}$$

BMR uses the corresponding target-margin gain and background deficit,

$$G = M_{\mathrm{atk}} - M_{\mathrm{bg}}, \qquad D = [-M_{\mathrm{bg}}]_+. \tag{17}$$

Consider a sample for which the background model does not predict the target class, i.e., $M_{\mathrm{bg}} < 0$. Then $D = -M_{\mathrm{bg}}$, and

$$M_{\mathrm{atk}} = M_{\mathrm{bg}} + G = G - D. \tag{18}$$

The attacked model therefore crosses the target decision boundary exactly when

$$M_{\mathrm{atk}} > 0 \iff G > D. \tag{19}$$

If a positive target margin of at least $\kappa > 0$ is required, the condition becomes

$$M_{\mathrm{atk}} \geq \kappa \iff G \geq D + \kappa. \tag{20}$$

This equivalence is a direct consequence of the definitions rather than a separate theoretical result. We use it to interpret the measurements: the background margin determines how much must be recovered, while G records the additional target margin obtained after introducing the malicious model.

The same identity connects the condition for individual samples to attack success. Among samples with a positive deficit,

$$\Pr\left(M_{\mathrm{atk}} > 0 \mid M_{\mathrm{bg}} < 0\right) = \Pr\left(G > D \mid M_{\mathrm{bg}} < 0\right). \tag{21}$$

Appendix C evaluates this relation directly on the actual merged models.

A.2. Reference State and Residual Contribution

The final BMR training procedure uses

$$\theta_{\mathrm{bg}} = \theta_0, \qquad \theta_{\mathrm{atk}} = \theta_0 + \alpha\delta_{\mathrm{adv}}. \tag{22}$$

The two states use the same triggered input and differ only by the scaled malicious task residual. Consequently,

$$G = M(\theta_0 + \alpha\delta_{\mathrm{adv}}, x^\tau) - M(\theta_0, x^\tau) \tag{23}$$

measures the change in target margin caused by the malicious residual at the sampled scale.

This differs from directly maximizing M_{atk}. An absolute-margin objective includes the margin already present in the pretrained reference, whereas the difference in Eq. (23) removes this term and focuses optimization on the additional margin introduced by the malicious task model.

A.3. Effect of Residual Scaling

The coefficient α varies the effective strength of the malicious task residual. Using the fixed pretrained reference, define

$$G_\alpha = M(\theta_0 + \alpha\delta_{\mathrm{adv}}, x^\tau) - M(\theta_0, x^\tau). \tag{24}$$

Away from class-tie points, the active non-target class remains locally fixed and the margin is differentiable. A first-order expansion around θ_0 gives

$$\begin{aligned} G_\alpha = \alpha \nabla_\theta M(\theta_0, x^\tau)^\top \delta_{\mathrm{adv}} \\ + O\left(\alpha^2 \|\delta_{\mathrm{adv}}\|_2^2\right). \end{aligned} \tag{25}$$

Thus, the available target-margin gain depends directly on the effective scale of the malicious residual. Sampling α exposes the same residual to different contribution strengths during training, rather than assuming a fixed downstream coefficient.

A.4. Optimization Behavior of the BMR Objective

Let

$$\bar{D} = \mathrm{sg}(D), \qquad a = \bar{D} + \epsilon. \tag{26}$$

In the active region $G > \epsilon$, the BMR objective can be written as

$$\mathcal{L}_{\mathrm{BMR}} = \log\left(1 + \frac{a}{G + \epsilon}\right). \tag{27}$$

Its derivative with respect to G is

$$\frac{\partial \mathcal{L}_{\mathrm{BMR}}}{\partial G} = -\frac{a}{(G+\epsilon)(G+\epsilon+a)} < 0. \tag{28}$$

Minimizing the objective therefore increases the target-margin gain.

For fixed G,

$$\frac{\partial}{\partial \bar{D}} \left| \frac{\partial \mathcal{L}_{\mathrm{BMR}}}{\partial G} \right| = \frac{1}{(G+\epsilon+a)^2} > 0. \tag{29}$$

Hence, a larger background deficit produces a stronger optimization pressure to increase G.

The stop-gradient operation keeps this deficit as a reference quantity during the update. With $\bar{D} = \mathrm{sg}(D)$,

$$\nabla_{\delta_{\mathrm{adv}}} \mathcal{L}_{\mathrm{BMR}} = \frac{\partial \mathcal{L}_{\mathrm{BMR}}}{\partial G} \nabla_{\delta_{\mathrm{adv}}} G, \tag{30}$$

while no gradient is propagated through the reference deficit. The optimization therefore acts through the target-margin contribution of the malicious residual.

A.5. Parameter Coupling Measure

For completeness, we provide the parameter-space coupling measure used in Fig. 1. For adversary task i and merging method m, let

$$\Delta_i = \theta_i - \theta_0 \tag{31}$$

denote the clean task residual of the adversary, and let

$$\Delta_{\mathrm{bg}}^{-i,m} = \theta_{\mathrm{bg}}^{-i,m} - \theta_0 \tag{32}$$

denote the benign background obtained by merging the remaining clean task models under merging method m. We measure their absolute global cosine similarity as

$$C_i^m = \left| \frac{\left\langle \mathrm{vec}(\Delta_i), \mathrm{vec}(\Delta_{\mathrm{bg}}^{-i,m}) \right\rangle}{\| \mathrm{vec}(\Delta_i) \|_2 \| \mathrm{vec}(\Delta_{\mathrm{bg}}^{-i,m}) \|_2} \right|. \tag{33}$$

This is the quantity reported as parameter coupling in Fig. 1. It measures global alignment between the clean task residual of the adversary and the benign merged background in parameter space. We use this measurement as a descriptive proxy in the motivation experiment; it is not interpreted as a causal measure of backdoor robustness.

Appendix B. Training Configuration

The final BMR method uses the shared pretrained model as the fixed reference: $\theta_{\mathrm{bg}} = \theta_0$. No synthetic parameter drift is added during training. At each malicious-training step, the residual coefficient is sampled as

$$\alpha \sim \mathcal{U}(0.2, 1), \tag{34}$$

and the attacked state is

$$\theta_{\mathrm{atk}} = \theta_0 + \alpha \delta_{\mathrm{adv}}. \tag{35}$$

The BMR loss is then evaluated using the two target margins defined in Section 4.1. Synthetic parameter drift is used only as an ablation in Section 5.9 and is not part of the final method.

TABLE 12. IMPLEMENTATION CONFIGURATION FOR THE VIT-B/32 EXPERIMENTS.

Component	Configuration
BMR training	
Residual coefficient	$\alpha \sim \mathcal{U}(0.2, 1)$
Trigger construction	
Trigger size	22×22 for 224×224 inputs
λ_{sig}	0.1
λ_{dor}	1
λ_{amp}	10^{-4}
λ_{tv}	10^{-3}

TABLE 13. FRACTION OF SAMPLES WITH $D > 0$ SATISFYING $G > D$, WITH THE CORRESPONDING ASR (%).

	Ordinary		Scale Var.		BMR	
Merger	$G > D$	**ASR**	$G > D$	**ASR**	$G > D$	**ASR**
TA	32.33	36.30	99.97	99.97	**100.00**	**100.00**
TIES	23.68	31.25	99.29	99.36	**100.00**	**100.00**
RegMean	16.39	22.96	88.38	89.45	**97.27**	**97.53**
AdaMerging	31.13	34.86	99.85	99.86	**100.00**	**100.00**
EMR	99.69	99.65	**100.00**	**100.00**	**100.00**	**100.00**
RegMean++	10.26	12.87	90.09	90.59	**98.18**	**98.28**

Appendix C. Sample-Level Target-Margin Analysis

Appendix A.1 shows that, for a triggered sample with $M_{\mathrm{bg}} < 0$, the attacked model predicts the target class exactly when $G > D$. We verify this relation on the six actual merged models.

For each merging algorithm, we construct matched clean and malicious merges using the same benign contributors and merging configuration. The malicious contributor is produced by Ordinary training, Scale Variation, or BMR. We evaluate all 9,900 triggered non-target CIFAR-100 test samples.

Figure 6 shows a clear progression from Ordinary training to Scale Variation and BMR. Varying the residual strength already moves many samples above $G = D$, while the BMR objective shifts the distribution further. The additional improvement is most visible under RegMean and RegMean++, where the 10th percentile of the target-margin gain remains challenging.

As shown in Table 13, the fraction satisfying $G > D$ closely tracks ASR across all six merging algorithms, as expected from Eq. (19). The two quantities are not identical because the former is conditioned on $D > 0$, whereas ASR is computed over the complete triggered non-target set. The value of this analysis is therefore the measured increase in G from Ordinary training to Scale Variation and BMR, rather than the identity itself.

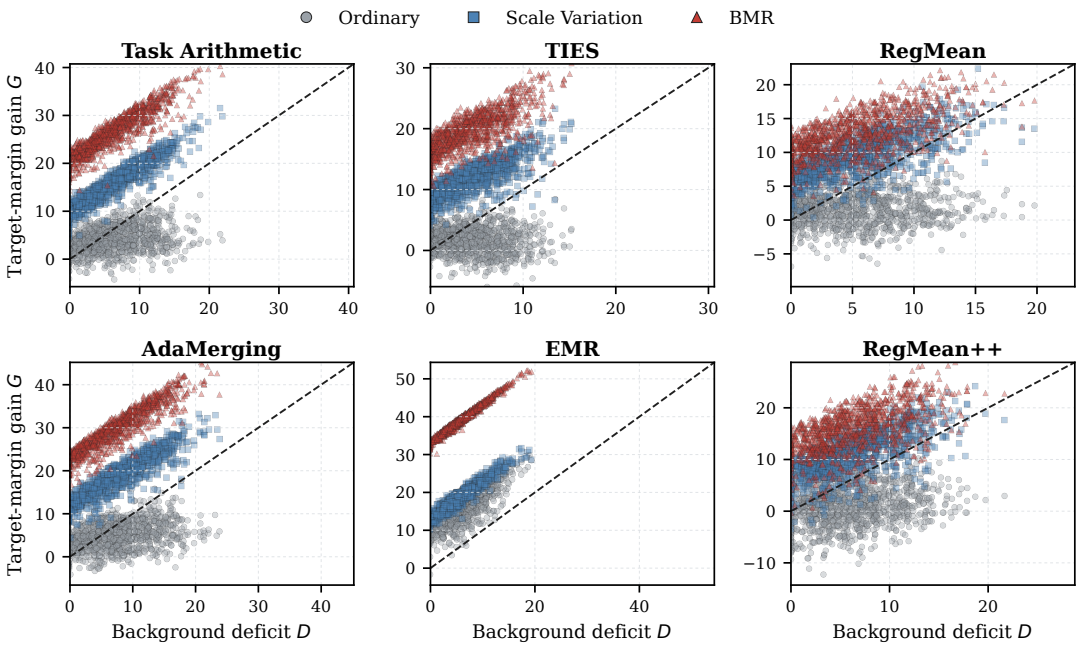


Figure 6. Sample-wise relation between background deficit D and target-margin gain G under actual merging. The dashed line marks $G = D$; for $D > 0$, points above the line satisfy the condition for target prediction.